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Lehrbuch der Algebra

Working English Translation Batch 07

Linear Substitutions and Inverse Substitutions

Translator's note

This working English translation is based on the cleaned Weber source currently in hand. It translates the opening of Weber's treatment of linear substitutions: the definition of a linear substitution, the law of composition, similarity substitutions, and the existence of inverse substitutions. I have preserved Weber's formulas and order of exposition while lightly normalizing punctuation and notation.

§37. Linear substitutions and their composition

One of the most effective means for forming groups, and at the same time one that leads to many applications, is given by linear substitutions and their composition. We have already encountered such linear substitutions several times, for example in the second section of the first volume on the occasion of the multiplication law for determinants, and then again in the eleventh section in connection with equivalent numbers.

By a *linear substitution* in n variables we understand a system of equations by which one system of n variables $y_1, y_2, \dots, y_n$ is expressed linearly in terms of another system $x_1, x_2, \dots, x_n$. According to the number of variables we distinguish unary, binary, ternary, quaternary substitutions, and in general we shall call the number of variables the *dimension* of the substitution. For the time being we restrict attention to homogeneous substitutions, so that, if the coefficients are denoted by $a_i^{(\kappa)}$, the substitution is represented by the system

$$y_\kappa = \sum_{i=1}^n a_i^{(\kappa)} x_i \quad (\kappa = 1, 2, \dots, n). \quad (1)$$

Very often the variables themselves are not important, so that such a substitution is sufficiently characterized by its coefficients $a_i^{(\kappa)}$. One sometimes uses a single letter, say A , to denote a substitution, setting

$$A = \begin{pmatrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_n^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_n^{(2)} \\ \vdots & \vdots & \ddots & \vdots \\ a_1^{(n)} & a_2^{(n)} & \cdots & a_n^{(n)} \end{pmatrix}. \quad (2)$$

We shall also write briefly $A = (a_i^{(\kappa)})$, and denote the determinant of the substitution coefficients by $|A|$. This determinant is always assumed to be nonzero. We also write the system symbolically as

$$(y_1, y_2, \dots, y_n) = A(x_1, x_2, \dots, x_n), \quad (3)$$

or, more briefly,

$$(y) = A(x). \quad (4)$$

The substitution

$$y_1 = x_1, \quad y_2 = x_2, \quad \dots, \quad y_n = x_n \quad (5)$$

or

$$J = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} \quad (6)$$

is called the *identity substitution*.

A substitution of the form

$$y_1 = \mu_1 x_1, \quad y_2 = \mu_2 x_2, \quad \dots, \quad y_n = \mu_n x_n \quad (7)$$

or

$$M = \begin{pmatrix} \mu_1 & 0 & \cdots & 0 \\ 0 & \mu_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mu_n \end{pmatrix} \quad (8)$$

will be called a *multiplicative substitution*, or simply a *multiplication*; the elements $\mu_1, \mu_2, \dots, \mu_n$ are called the *multipliers*.

Now suppose that we have a second linear substitution of the same dimension n , introducing a new system of variables z :

$$z_i = \sum_{h=1}^n b_h^{(i)} y_h, \quad (z) = B(y). \quad (9)$$

If we substitute for y the values from (1), then we obtain the z expressed in terms of the x by means of a new linear substitution E , which we denote by

$$z = E(x) = BA(x), \quad (10)$$

and the substitution $E = BA$ is said to be *composed* from B and A . One must, however, distinguish between AB and BA . Two substitutions A, B having the special property $AB = BA$ are called *interchangeable* or *commutative*.

The coefficients of E are obtained by inserting the expressions (1) into (9):

$$z_\kappa = \sum_h e_h^{(\kappa)} x_h, \quad e_h^{(\kappa)} = \sum_{i=1}^n b_i^{(\kappa)} a_h^{(i)}. \quad (11)$$

These formulas are exactly the same as those used earlier for the multiplication of determinants. Thus we may express the composition law of substitutions as follows.

1. To form the substitution BA composed from B and A , one proceeds exactly as if the determinants $|B|$ and $|A|$ were to be multiplied by the determinant multiplication rule. In order to obtain the elements of one row, one multiplies the elements of a row of the first factor by the corresponding elements of the columns of the second factor and then adds.

From this rule there follows immediately:

2. The determinant of a composed substitution is equal to the product of the determinants of its components.

To compose three substitutions C, B, A of the same dimension, we insert the expressions (10) for z into a further linear substitution

$$(u) = C(z). \quad (12)$$

Then the variables u are expressed in terms of the x by

$$(u) = CBA(x). \quad (13)$$

Since it is evidently immaterial whether one first introduces (10) into (12), or first expresses z by y and then y by x , one obtains for this composition the associative law

$$C(BA) = (CB)A = CBA. \quad (14)$$

The same may of course be verified directly from (11). Thus:

3. For the composition of linear substitutions, the associative law holds, but in general not the commutative law.

A multiplication in which all multipliers are equal,

$$N = \begin{pmatrix} \nu & 0 & \cdots & 0 \\ 0 & \nu & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \nu \end{pmatrix},$$

will be called a *similarity substitution*. Two substitutions that are obtained from one another by composition with a similarity substitution, such as A and AN or A and NA , are called *similar*. From the composition law one sees immediately:

4. A similarity substitution is commutative with every substitution A of the same dimension; two similarity substitutions, composed with one another, again yield a similarity substitution; and two substitutions that are each similar to a third are also similar to one another.

Equally immediate is the statement:

5. Composition with the identity substitution J leaves every substitution unchanged.

The substitution J therefore plays the role of a unit in composition and may be omitted wherever it occurs composed with other substitutions.

We have also the following theorem.

6. To every substitution A there exists one and only one inverse substitution A^{-1} satisfying

$$AA^{-1} = A^{-1}A = J. \quad (15)$$

This last statement follows from the fundamental formulas of determinant theory if one determines the elements $\alpha_h^{(\kappa)}$ of A^{-1} from the linear equations

$$\sum_i a_i^{(h)} \alpha_\kappa^{(i)} = \begin{cases} 0, & h \neq \kappa, \\ 1, & h = \kappa, \end{cases} \quad (16)$$

from which, if $A_\kappa^{(h)}$ denotes the minor of $|A|$, one obtains

$$|A| \alpha_h^{(\kappa)} = A_\kappa^{(h)}. \quad (17)$$

These in turn imply the system

$$\sum_i \alpha_i^{(h)} a_{\kappa}^{(i)} = \begin{cases} 0, & h \neq \kappa, \\ 1, & h = \kappa. \end{cases} \quad (18)$$

The inverse substitution of A is nothing other than the solution of the system of equations defining the direct substitution A , so that from

$$(y) = A(x)$$

there follows

$$(x) = A^{-1}(y). \quad (19)$$

For the composition of inverse substitutions one obtains from $ABB^{-1}A^{-1} = J$ the formula

$$(AB)^{-1} = B^{-1}A^{-1}. \quad (20)$$

If A, B, C are three substitutions, then composition with A^{-1} shows from either of the equations

$$AB = AC, \quad BA = CA$$

that necessarily $B = C$. Hence the characteristic properties required for a group are satisfied by the composition of substitutions, and thus:

The totality of all substitutions of a fixed dimension forms an (infinite) group.